

Early Prediction of Diabetes Disease & Classification of Algorithms Using Machine Learning Approach

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ABSTRACT

Diabetes is a global health issues that usually prolonged in a patient for an entire life. Diabetes effect people of all age groups. Development of technology is a non-conventional approach to predict diabetes and provide accurate results with better efficiency. Most of the researches have been carried out to predict the diabetes in which most of the researchers have used Pima Indian dataset. In this paper, authors build a Framework that can estimate with maximum precision, probability of diabetes in Patients. In this paper, authors build a Framework that can estimate with maximum precision, probability of diabetes in Patients. As a result, authors would like to use Machine Learning approach such as Decision Trees, SVM and Naive Bayes to detect diabetes in early phase. The Pima Indian Diabetes that are obtained from UCI library test to measure although these algorithms are used to predict diabetes in order to save time and get accurate results.

Keywords: Machine Learning, Diabetes, Precision, Classification, SVM, Naïve Bayes, Decision Tree

1. Introduction

Hypertension is an acute symptom that gives rise to high blood sugar levels called as diabetes which has adverse effects on human health. Diabetes occurs when there is a huge deviation of body sugar levels from the normal, either too high or too low. Diabetes can result in multiple organ failures like Kidneys, liver, heart and other organs which can ultimately result in the death of people [1]. Hypertension affects various areas of the body including the lungs, kidneys, heart, and nerves so on. According to Williams' endocrinology report, more than 382 million individuals worldwide or all across the universe were diabetic in 2013. Every year, hypertension disease kills a large number of people in both developing and an advanced country around the world. Diabetes patient requires daily attention and any negligence in the treatment can prove fatal. To provide better health care of diabetes patient in the prediction the glucose levels, Machine Learning approaches have been suggested and recommended. Machine Learning approach can play a profound role in early diabetes detection, prediction and preventive measures in diabetic patients so as to improve the health care such as SVM, NB, ANN and other algorithms to recognize patterns in data Machine Learning scientific knowledge learning methods. We can achieve highly correct estimates using these approaches [2]. Diabetes is a fast illness that causes people when the insulin hormone is produced by the pancreas and the cells of the body cannot use it or when the kidneys fail to produce sufficient insulin [3]. In 2012 five million deaths were caused due to the diabetes. The statistics released by the World Health Organization (WHO) also proved to be 9 percent of adults and older people had diabetes in 2014[4]. Diabetes mellitus (DM) is a prolonged non-communicable illness. The disease has been carefully monitored by the WHO and the World Health Federation (WHF) since the number of diabetes has steadily increased. Hypertension also known as diabetes is a set of cardiovascular problem faced by diabetes over a long period of time. Side effects of high glucose include excessive urination every time feeling thirsty and stomach cramps [5]. If diabetics are not treated properly then hypertension can cause serious health problems such as hypoglycemia, hypotonic hypertensive status or even lead to death [6]. It might lead to major health problems such as cardiovascular disease, stroke, kidney disease, pressure ulcers, forehead complications, etc. Different classifiers are mostly used during the health industry for sorting data into various categories based to many limitations unlike a particular classification model. Obesity is a serious disease which really damages the body processes the hormone cortisol which causes metabolic pathway to become irregular and blood glucose levels to rise. High blood pressure is considered a severe health burden in which the glucose level in the blood cannot be regulated. Blood pressure is affected by a combination of conditions such as age, weight, genetic factors and insulin but the most important factor to remember is sugar concentration. The only way to avoid problems is to detect the problem earlier stage [7]. Many researchers have conducted simulations to identify diabetes. As research have shown Machine Learning datasets such as SVM, NB, DT and others can be used to classify diabetes with different methods and so on. Using various classification methods of Machine Learning approaches and so on .As studies have shown that Machine Learning algorithms can accurately diagnose diseases [8-10].. The ability to manage a large amount of data gives machine learning algorithms such power of combining data from a variety of sources and incorporating the study's metadata [11]. The PIDD dataset is used to test NB, SVM, and DT results for controlling diabetes in patients. On different levels a test quality of all three algorithms is compared and they all achieve better precision [12]. The research paper contributes the following contributions which are given below:

- In this paper authors has done the complete classification of diabetic prediction using Machine Learning approach such as NB, SVM and Decision tree.
- Authors proposed a Framework for Diabetes prediction.
- The Classification of accuracy is compared and finds the best Machine Learning algorithm.

The majority of a paper is well structured as the follow Segment 2 explains about the Literature Survey of various classification techniques for prediction of diabetes. Segment 3 represents the Framework. Segment 4 describes the methodology and brief discussion of data set. Segment 5 discusses about the various diabetic predictions and comparisons. Segment 6 discusses about result and finally the Segment 7 determines the conclusion and future work.

2. Literature Survey

A function of an Algorithm and Classification ensemble Machine Learning approach is discussed. Depending on diabetes risk factors the J48 decision tree was used to identify hypertension whether the Patient is diabetic or non-diabetic [13]. The research's finding shows that Ad boost Machine Learning ensemble technique outclasses bagging and a J48 decision tree in terms of efficiency. It developed a glucose predictive model with the primary goal of predicting whether or not a patient will develop diabetes at a given age. The conceptual model is widely used in machine learning and employs decision trees to solve problems to implement it [14]. The observed results are accurate because the developed method performs well in detecting diabetic events at a specific age with greater accuracy by using a Prediction Model [15]. For testing and evaluating the dataset for estimation support vector was used. High blood pressure collection of data sourced from UCI database was used to examine disease [16-17]. Through Program is designed and we were able to obtain the best data. By reducing the time needed to generate a dataset reliability can be extremely high. It tends to be expensive for diabetes prediction a hypertension data sampling model with two sub-modules [18]. In the first partition an Artificial Neural Network is used and in the second partition. Fasting Blood Sugar is used for decision tree and is used to identify an effects of hypertension on the health of patients [19]. It can also be applied to the algorithm which can classify the risk of diabetic mellitus [20]. The author used four well-known Machine Learning classification methods to achieve the goal DT, ANN and Naive Bayes. Methods like classifiers are used to increase the reliability to build models. The Random Forest algorithm yields the best outcomes of all the algorithms and tested according to the results of the research. The Machine learning approach is critical for predicting a number of medical databases, including hypertension data [21]. The goal of this research is to develop a structure in a medicine device that provides machine learning to predict hypertension using significant features that are mainly associated to the illness [22]. Although data is one of the most essential issues of classification process, machine learning approaches are entirely reliant on it. When data is obtained from various sources in a raw format, there is a risk of several variations, which the model might not be able to manage. As a result, pre-processing is proposed to reduce all variations and create a pure set of data [23].

3. Frame work

In this section proposed Framework is associated with the diabetes prediction in which author uses the Data base of people with diabetes as shown in Fig.1

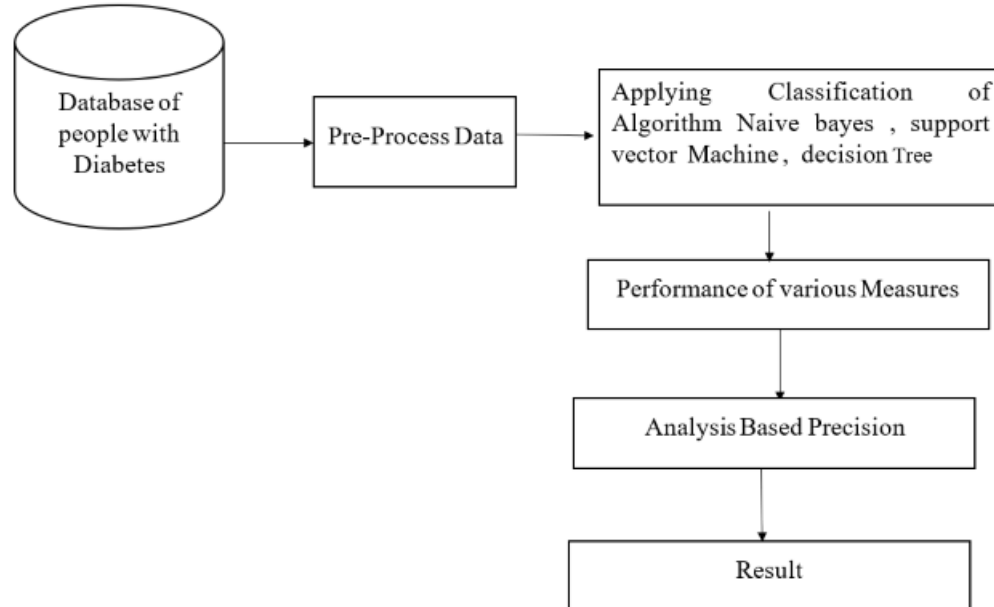


Fig.1 Proposed Frame Work of Diabetes Prediction

Author take the database of people with diabetes then author preprocess the data by applying the different classification of Algorithm such as NB, SVM, DT. After that author do performance on various measures by applied the analysis based precision and finally the author gets the result for predicting the diabetes.

4. Methodology

Different research on diabetic disease has been developed. Previously, several researchers conducted various studies in medical centers. Income was also spent by some of those researchers on diabetic patients, as it was a set of problems in the old age of research, which was performed only in health facilities rather than using a different surface like machine learning.

Machine Learning Approach: Machine Learning Algorithm will create the design of our data set based on the problem. In this paper the author has used three algorithms such as NB Algorithm, SVM and DT.

SVM: It is supervised machine learning. It consists of class training sample. SVM is used to find the best straight line between the two classes for the good conclusion straight line should not be positioned nearer to the data set of the other. Group of straight line is far away from the data set in each group should be chosen the area that are nearest to SVM are positioned at the classify margin. The researcher's precision is evaluated Using the WEKA Software. The SVM identifies the best distance matrix by growing the gap between the two possible outcomes. In a mathematical sense Author will enhance the distance between the straight lines, as shown below;

$$\frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i (w^T x_i - b)) + \lambda ||w||^2$$

SVM should be correctly classified all y (j) which means $(wT xi + b) \geq 1$. The efficiency of SVM algorithm for diabetic diagnostic was evaluated [24-25]. Table 1 illustrates how to use the Confusion Matrix:

Table 1: Matrix of Confusion by Support Vector Machine

	A	B
A Test positive	499	0
B Test Negative	267	0

Naïve Bayes Algorithm: Naïve Bayes Algorithm classifies the data set using the Bayes rule of probability. It is optimization methodology built on the concept that all the attributes are unrelated and separate to one another. It specifies also the classification of one feature in a class that has no impact on the status of another feature of the same class. It's a machine learning algorithm that's supervised. The classification is done on the given probability of an equation. It works well for the knowledge that has the challenge with balance and missing value [26]. By using the Naïve Bayes theorem it can be calculated on the basis of probability a/y can be derived from $Pr(a)$, $pr(b)$ and $pr(y/a)$ [27].

Then

$$Pr(Pr(a|y) = (Pr(y|a)) | pr(y) \quad (2)$$

Where

$Pr(a|y)$ = condition probability of class.

$Pr(y|a)$ = medium of class probability.

$Pr(a)$ = class a probability of being correct.

$Pr(y)$ = it is the accuracy of chance for a performance of NB Algorithm using confusion of the matrix.

Table 2: Confusion of the matrix by NB

	A	B
A Test positive	421	77
B Test Negative	103	163

Decision Tree Algorithm: DT Algorithm is a classification algorithm using the ensemble learning concept it will classify the data set. It is a supervised Machine Learning approach for solving the classification problem. The aim of a decision tree in the research is to forecasting the specific class using DT at a specific data set. It uses both the node and internode for the recognition and diagnosis. The decision is performed by the roots of the tree in various ways of the attributes. The tree's root may have two or more branches and the leaf nodes show the classification. In every step decision tree chooses the highest knowledge gain of the attributes used to select the nodes of every tree [28]. Table 3: displays the uncertainty matrix in the decision tree

Table 3: Confusion of matrix by decision tree

	A	B
A Test positive	406	92
B Test Negative	107	159

Data set description : This work is done under the experiment performance of WEKA. WEKA is a Specific Software advanced by the university of Waikato that involves the selection of Machine Learning approach for data classification process such as classification, regression, clustering etc. [29-30]. The significant advantage of the WEKA is that it can be customized to meet the specific need. The primary aim of this research is to use the WEKA tools to predict diabetes in patients by using the data base of (PIDD) Pima India Diabetes Data Set. In Table 4: An author's shows the description of data set.

Table 4: Data set a brief description

Database	Number of attributes	Number of instances
PIDD	7	767

Pima India Diabetes Data set

A suggested technique put into practice when it comes to diabetes data set namely PIDD which authors have taken from UCI Repository [31]. The data set includes the clinical data of 767 female patients. It also includes the numerical values of 7 attributes where as one class is treated as '0' for the test negative for the disease of diabetic and also with the of class '1' is tested as positive for diabetic disease. It defined the description of data set by table 4, 5 which represents the explanation of attributes.

3.1 Accuracy Measurements

SVM, NB and DT algorithm authors in a research work. Cross validation is of ten folds. Recall, Accuracy, F-measure, precision and ROC can be used for the classification of the work is shown in Table 5:

Table 5: Describes the precision measure given below.

S. No.	Measures	Definition	Formula
1	Certainty	The algorithm's reliability in measuring situations is determined by its precision.	$A = (TPR + TM) / (\text{Total no of samples})$
2	precision	It is measured in accuracy in the form of precision.	$PR = TPR / (TPR + CP)$
3	Recall	Recall is used to assess the correctness or reliability of a classification model.	$R = TPR / (TP + CM)$
4	f- measure	The weighting factor of accuracy is called F-Measure.	$F = 2 * (P * R) / (P + R)$
5	ROC	To evaluate the utility of experiments, ROC (Receiver Operating Curve) curves are used.	

Table 6: Performance and Classifications of algorithms

Algorithm	Precision	Recall	Measure	Accuracy	ROC
Naïve Bayes	0.757	0.761	0.758	74.28	0.817
Support Vector Machine	0.422	0.649	0.511	63.10	0.549
Decision Tree	0.733	0.736	0.734	71.81	0.749

The output of the respective classification algorithm in terms of Accuracy, Recall, f-measure, ROC is described in above Table 6: on the basis of instances in Table 7: the output classifier is specified.

Where TP is known as true positive TN is known as true negative FP is known as false positive FN is known as false negative. Accuracy, Recall, f measure, ROC values are used to analyze the quality of the corresponding classifier on the basis of correctly classified while Table 7: shows the accuracy of the model on the basis of unclassified instances.

Table 7: Classification instance performance

Total number of instances	Classifications of Algorithms	Categorized Correctly	Categorized Wrongly
767	Naive Bayes	585	181
	Support Vector Machine	499	267
	Decision Tree	566	200

4. Various Diabetic Prediction Methodologies Comparison

The Table 8: below provides a description of the different diabetes prediction approaches. It also makes suggestions that can be applied in the future.

Table 8: Shows Diabetic Prediction Methodologies Comparison

Ref. No.	Objective	Method used	Result	Advantages	Disadvantages	Recommendation
32	In the proposed system author uses diabetes prediction with various classification such as Navies Bayes, SVM, Decision tree	Machine learning, Ad boost	Ad boost decision support system has an precision of 80.72%	It uses a general data base with 768 instances and 9 attributes as well as local database for validation.	It is indeed just to predict more in female women.	It is indeed possible incorporated with certain type of classifier's K Nearest for eg ANN to know further, join a community Quantitative measurement
33	Author may use six machine learning .It is a tool for prediction of diabetes.	Machine learning, NB K- nearest neighbor, SVM ,DT	The Methods were compared on Pima India Diabetes data set .The precision of the KNN; SVM algorithm is	This method compares the performance of various machine learning techniques and calculates which algorithm is superior.	One of the drawbacks of this method is that some of the input variables were	

			77%which is better than other algorithm.		missing from the data set they used.	
34	The current technique utilizes supervised machine learning algorithm.	SVM,KNN ,Naïve Bayes ,ANN	Mostly on the Pima Indian diabetes data achieves a high precision of the other entire algorithm using min max scaling.	They are able to enhance diabetes predictive performance by using Min max Scaling process.	The detection Precision of ANN was poorer	Other classification Such as Decision tree, K Means may be used to execute it.
35	To predict hypertension the classification model is based on 4 supervised ML and the WEKA tool.	NB,SVM, Random Forest CART	When it comes to the necessary consequence the SVM outperforms all other classifier's with a score of 79%	Model that has been proposed and constructed utilizing WEKA instrument an Experiment at the end result could be in observational.		
36	The advanced computer which was used to predict diabetes disease can affect the people	SVM ,Navies Bayes,j48	J48 has a good performance superior to other technique i.e. use of 0.8%.SVM is a collection of Nodes .Navies Bayes kept track of everything at the lowest worth.	They created an Aneroid app that is user-friendly and offers illness advice and data.		
37,38	To predict insulin proposed Framework employers four Machine learning techniques for the risk factor of early prediction of diabetes.	SVM,C45,Decision tree,Naives Bayes, KNN	Highest precision is 73%on decision tree than other algorithm.	It can be used in different technology for prediction and comparing the precision and decide which technique is good.		

5. Result and Discussion

Table 6: represents the Algorithm classification efficiency varies widely based on different metrics. From Table 6: shown that NB has the highest accuracy, in the contrast to other classification models. In contrast to other classification models, the Machine Learning approach can predict the risk of insulin with greater accuracy. Analysis and results showed that the Naïve Bayes algorithm yields the better result when compared with Support Vector Machine and Decision tree. The performance can be plotted in the graph via Fig. 2 and Fig. 3, Fig4 represents the all classification algorithms ROC location is represented. Three classification algorithms are used in this study to generate a framework that aids in the early-stage prediction of Diabetic using significant features related to this illness. The algorithms are Naïve Bayes, SVM, DT. These algorithms are used from the dataset of PIDD obtained from UCI database.

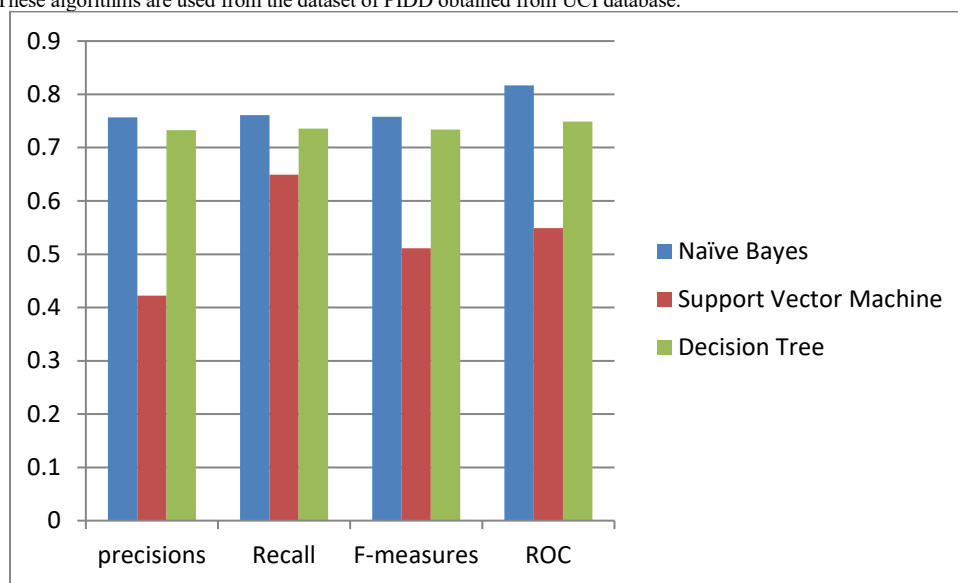


Fig. 2 represents the performance of the graph for different Algorithms

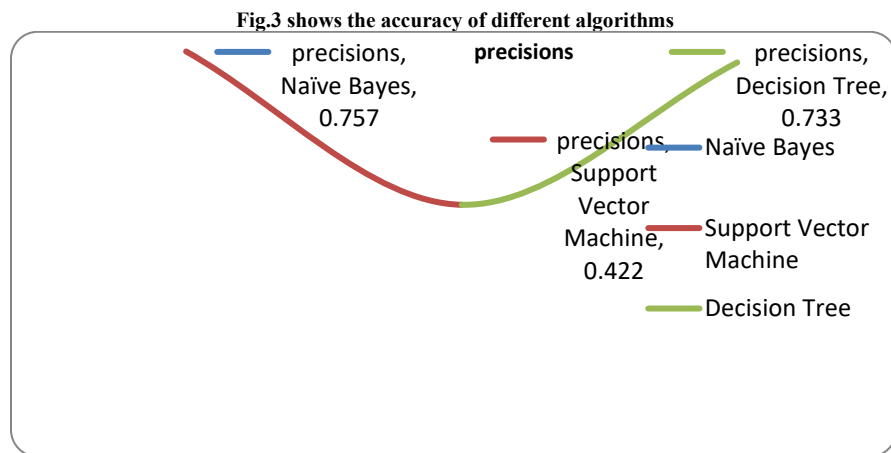
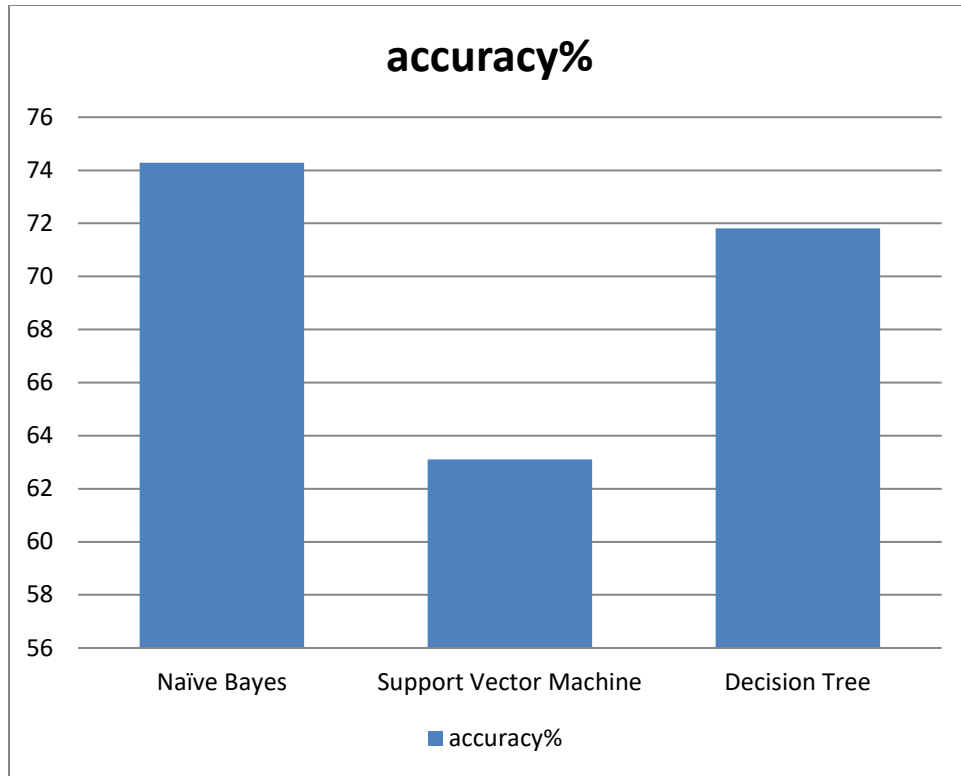


Fig. 4 Represents the ROC Curve for algorithms.

Table 7: defines classification behavior based on instances which have already been classified. From the information, to the few defined examples accuracy is computed from a number of instances the performance of each algorithm is measured in terms of Correct Classified Cases and Incorrectly Classified Cases. On the basis of categorized cases Fig. 4 depicts the graphical efficiency of all classification algorithms. From Table 6: and Table 7: author analyze that the Naïve Bayes classification algorithm gives best result than other algorithms. The Experiment also shows that Naive Bayes has a accuracy of 74.28 percent compared to other Algorithms.

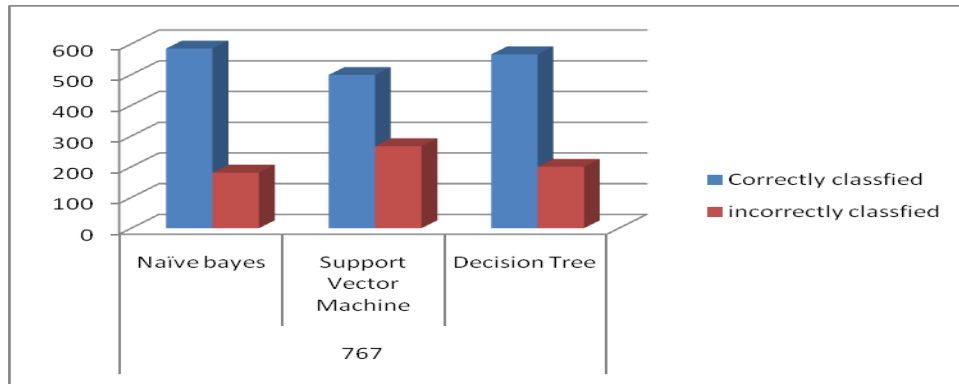


Fig .5 Represents the classified of instance

7. Conclusion and Future work

Early phase detection in diabetes is a serious research problem and a potential challenge for researchers in the field of human health care development. Diabetes is an actual harmful disease and early detection is always a challenge. This research employs machine learning classification algorithms to create a model is capable of overcoming all problems and is useful in the early prediction of diabetes disease. In this research structured efforts are made to design a Framework that can predict diabetes. The 3 Machine Learning approach classification algorithms are analyzed, tested on different measures as part of this research. The PIDD is used in the experiments. Using the NB classification algorithm, the experimental results show that a built method is adequate with an accuracy of 74.28 percent. The developed framework, along with the Machine Learning different classifiers used could be used to identify or diagnose other diseases in the future. For diabetes research, the study could be extended and improved as well as several other Machine Learning approaches and also authors have planned to do classification for other algorithms with missing data.

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